A Jose, Justin Gabriel D. **Object-Oriented Programming**

BSIT 2-1 **Midterm Activity #2 - Operators**

PART I:

1. Create a Java program using bitwise operators that will produce the output: true, 10, false, 15.

Act2PARTIno1.java > ...

```java
public class Act2PARTIno1 {
    Run | Debug
    public static void main(String[] args) {
        int a = 11;
        int b = 14;

        System.out.println(((a & b) == 10) + ", " + (a & b) + ", " + (a == b) + ", " + (a | b));
    }
}
```

```
PS C:\Users\Justin A Jose\COLLEGE\COLLEGE FILES 2ND YEAR\Java Programs> cd "c:\Users\Justin
A Jose\COLLEGE\COLLEGE FILES 2ND YEAR\Java Programs\" ; if ($?) { javac Act2PARTIno1.java }
; if ($?) { java Act2PARTIno1 }
true, 10, false, 15
```

2. Create a Java program using the ternary operator that will produce the output: true, false, 25, 30.

Act2PARTIno2.java > ...

```java
public class Act2PARTIno2 {
    Run | Debug
    public static void main(String[] args) {
        int a = 25;
        int b = 30;
        int min = (b>a)?a:b;
        int max = (a>b)?a:b;

        System.out.println((min <= 50) + ", " + (max > 50) + ", " + ((a != 25)?0:25) + ", " + ((a == 30)?0:30));
    }
}
```

```
PS C:\Users\Justin A Jose\COLLEGE\COLLEGE FILES 2ND YEAR\Java Programs> cd "c:\Users\Justin
A Jose\COLLEGE\COLLEGE FILES 2ND YEAR\Java Programs\" ; if ($?) { javac Act2PARTIno2.java }
; if ($?) { java Act2PARTIno2 }
true, false, 25, 30
```

3. Create a Java program using assignment operators that will produce the output: Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.

Act2PARTIno3.java > ...

```java
public class Act2PARTIno3 {
    Run | Debug
    public static void main(String[] args) {
        int add = 60;
        add += add;

        int subt = 200;
        subt -= 130;

        int multi = 3;
        multi *= 48;

        int div = 592;
        div /= 37;

        System.out.println("Addition: " + add + ", Subtraction: " + subt + ", Multiplication: " + multi + ", Division: " + div);
    }
}
```

```
PS C:\Users\Justin A Jose\COLLEGE\COLLEGE FILES 2ND YEAR\Java Programs> cd "c:\Users\Justin
A Jose\COLLEGE\COLLEGE FILES 2ND YEAR\Java Programs\" ; if ($?) { javac Act2PARTIno3.java }
; if ($?) { java Act2PARTIno3 }
Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16
```

PART II:

Program Requirements:

1. Use at least five (5) different operators from the following categories:
   - Arithmetic operators (+, -, *, /, %)
   - Relational operators (>, <, >=, <=, ==, !=)
   - Logical operators (&&, ||, !)
   - Unary operators (++, --)
   - Assignment operators (+=, -=, *=, /=)
   - Bitwise operators (&, |, ^, <<, >>)

2. The program must produce exactly six (6) outputs only.

3. At least one output must include a tricky increment or decrement case, such as:
   - a++ + ++a
   - --b + b++
   - or any similar expression that modifies the variable within the same statement.

4. Do NOT reuse:
   - Any variable values
   - Any variable names
   - Any logic patterns
     from Part I.

5. The output must contain:
   - At least two (2) boolean results
   - At least four (4) numeric results

Act2PARTII.java > Act2PARTII

```java
public class Act2PARTII {
    Run | Debug
    public static void main(String[] args) {

        int alpha = 20;
        int bravo = 4;
        System.out.println(alpha - bravo);

        int charlie = 15;
        System.out.println(charlie < 10);

        boolean delta = true;
        boolean echo = false;
        System.out.println(delta && echo);

        int foxtrot = 5;
        System.out.println(foxtrot++ + --foxtrot);

        int golf = 3;
        golf += 27;
        System.out.println(golf);

        int hotel = 6;
        System.out.println(hotel >> 1);
    }
}
```

```
PS C:\Users\Justin A Jose\COLLEGE\COLLEGE FILES 2ND YEAR\Java Programs> cd "c:\Users\Justin A Jose\COLLEGE\COLLEGE FILES 2ND YEAR\Java Programs\" ; if ($?) { javac Act2PARTII.java } ; if ($?) { java Act2PARTII }
16
false
false
10
30
3
```